irritating, especially when operated in large numbers for delivery or monitoring purposes. Continuous drone operations in populated areas may contribute to acoustic pollution, affecting both human comfort and animal behavior. Manufacturers are therefore focusing on aerodynamic propeller design, noise-dampening materials, and optimized flight control algorithms to reduce acoustic emissions and ensure compliance with urban noise standards.

From a sustainability perspective, the manufacturing and disposal of drones pose additional environmental challenges. Most drones are constructed from composite materials, metals, and electronic components that are difficult to recycle. The widespread use of lithium-ion and lithium-polymer batteries, while energy-efficient, contributes to the growing problem of electronic waste and hazardous chemical disposal. Developing biodegradable materials, recyclable components, and eco-friendly battery alternatives such as solid-state or hydrogen-based systems could significantly reduce the environmental impact of future UAV production.

Furthermore, the increasing reliance on drones for various industries also raises concerns about their carbon footprint during production, operation, and maintenance. Although electric propulsion is relatively clean, the energy used to charge large drone fleets, especially from non-renewable sources, can offset some of their environmental benefits. Integrating renewable energy-based charging systems, solar-assisted operations, and sustainable logistics practices can help balance technological advancement with ecological responsibility.

7.6.1 Drones impact on wildlife and ecosystems

While drones have become valuable tools in environmental monitoring, wildlife conservation, and ecological research, their increasing presence in natural habitats has raised concerns about unintended ecological